

Exact Traveling-Wave Solution Families of a Higher-Order Gain–Loss Nonlinear Schrödinger-Type Equation via the iB-Function (BDKm) Ansatz: A Combinatorial Solution Catalog and a Benchmark Design for Physics-Informed Machine Learning

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Abstract

We study a fourth-order nonlinear evolution equation that couples group-velocity dispersion, third- and fourth-order dispersion, saturable Kerr-type self-phase modulation, and gain/loss with finite gain-bandwidth, a model relevant to pulse propagation in dispersive, dissipative, gain-saturated media such as doped fiber amplifiers and mode-locked lasers. Using the traveling-wave reduction and the implicit Bogning function (iB-function) ansatz $A(\xi) = a J_{n,m}(\xi)$ with $J_{n,m}(\xi) = \sinh^m(\xi)/\cosh^n(\xi)$, we derive the fully expanded nine-term-times-two (“Range”) algebraic equation obtained after coefficient substitution, and we carry out a complete combinatorial classification of the index pairs (n, m) for which the two blocks of the Range equation can share common J -indices. This classification yields a restricted field of 49 rational half-integer pairs $(n, m) \in \{-3/2, -1, -1/2, 0, 1/2, 1, 3/2\}^2$, for which we tabulate the fully expanded coefficients $C_1, \dots, C_9$ in closed form in terms of the eight physical parameters of the model. Beyond the analytical contribution, we propose and specify a concrete methodology for using this parametrized exact-solution catalog as a benchmark and training-data generator for physics-informed neural networks (PINNs), extending existing PINN studies of the nonlinear Schrödinger hierarchy – which typically validate against one or two closed-form solitons – to a systematically generated, 49-branch family with known ground truth across the full physical parameter set. We give the explicit forward-problem and inverse-problem (parameter-discovery) loss functionals for this equation, work through one representative branch ($n = m = 1$) in full detail, and outline a generalization-testing protocol (training on a subset of (n, m) branches and evaluating on held-out branches) that is not available with a single hand-picked soliton solution. This paper is presented as a methodological framework and worked derivation; the numerical PINN experiments it specifies are proposed future work rather than results reported here.

Keywords: higher-order nonlinear Schrödinger equation; gain-saturated dispersive media; traveling-wave ansatz; iB-functions; exact solutions; physics-informed neural networks; benchmark datasets; parameter discovery

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1 Introduction

Nonlinear evolution equations in the nonlinear Schrödinger (NLS) hierarchy govern pulse propagation in a wide range of dispersive-nonlinear media, from optical fibers to Bose–Einstein condensates and deep-water waves. When higher-order dispersion, gain, loss, and saturable nonlinearity are all retained, the resulting model equation becomes analytically intractable by inverse scattering in general, and one instead seeks particular closed-form traveling-wave solutions through structured

ansätze (tanh-, sech-, Jacobi-elliptic-, and, more recently, “implicit Bogning function” or iB-function methods). These ansatz methods are algorithmic: a trial solution is substituted into the governing equation, the result is expanded in a basis of elementary functions, and the coefficients of each basis element are matched to zero, producing a system of algebraic constraints on the physical and wave parameters [6, 7].

Independently, over the last several years physics-informed neural networks (PINNs) [1] have become a standard tool for both the forward problem (learning a solution field from the governing PDE and sparse data) and the inverse problem (recovering unknown physical coefficients from field data) for equations in the NLS hierarchy. PINNs have been applied to the standard NLS equation, the third-order (Hirota) equation [2], higher-order NLS equations with self-steepening and Raman terms [3], equations with external potentials [4], and adaptive-sampling variants that improve accuracy on strongly localized structures such as rogue waves [5]. A recurring feature of this literature is that PINNs are trained and validated against a small number of hand-selected closed-form solutions (one soliton, two solitons, a breather, a rogue wave), each with its own fixed parameter values.

This paper makes two contributions. First, we present a complete and self-contained derivation of the traveling-wave reduction of a fourth-order gain-loss NLS-type equation using the iB-function ansatz $J_{n,m}(\xi) = \sinh^m(\xi)/\cosh^n(\xi)$, including a systematic combinatorial classification of which index pairs (n, m) can produce internally consistent solutions once the ansatz is substituted and the resulting expression is expanded over the saturable-nonlinearity denominator. This produces a *restricted field* of 49 admissible half-integer index pairs, each with fully expanded, closed-form coefficients in terms of the eight physical parameters of the model $(\beta_2, \beta_3, \beta_4, g_0, T_2, \alpha, \alpha_2, \gamma, \Gamma)$ and the two wave parameters λ_1, λ_2 .

Second, we argue – and specify in reproducible detail – that this kind of systematically generated, parametrized solution family is directly useful as a benchmark and training-data generator for PINNs applied to higher-order gain-saturated evolution equations, addressing a real gap in the existing PINN-for-NLSE literature: existing studies validate against isolated solutions rather than families, so they cannot test whether a learned solver or a parameter-discovery network generalizes across a structured solution space. We give the explicit PINN forward and inverse loss functionals for this equation (Section 9), work through the $(n, m) = (1, 1)$ branch as a concrete worked example, and specify a generalization-testing protocol. We do not report numerical PINN results in this paper; the machine-learning section is a methodological proposal intended to make the benchmark immediately usable by follow-up numerical work.

2 Governing Equation

We consider the fourth-order nonlinear evolution equation

$$\frac{\partial A}{\partial z} + \frac{i}{2}(\beta_2 + ig_0 T_2^2) \frac{\partial^2 A}{\partial t^2} + \frac{\beta_3}{6} \frac{\partial^3 A}{\partial t^3} + \frac{i\beta_4}{24} \frac{\partial^4 A}{\partial t^4} + i\left(\gamma + i\frac{\alpha_2}{2}\right) \frac{|A|^2 A}{1 + \Gamma|A|^2} + \frac{1}{2}(g_0 - \alpha)A = 0, \quad (1)$$

where $A(z, t)$ is the complex envelope, z is the propagation coordinate, t is the retarded time, $\beta_2, \beta_3, \beta_4$ are the second-, third- and fourth-order dispersion coefficients, g_0 is the small-signal gain, T_2 is the gain-medium dipole relaxation time (setting the parabolic gain bandwidth), α is the linear loss, α_2 is a two-photon-type loss coefficient, γ is the self-phase modulation (Kerr) coefficient, and Γ is the nonlinearity saturation parameter. Equation (1) reduces to the cubic NLS equation when $\beta_3 = \beta_4 = g_0 = \alpha = \alpha_2 = \Gamma = 0$, and to the Hirota/third-order NLS equation when $\beta_4 = g_0 = \alpha_2 = \Gamma = 0$.

3 Traveling-Wave Reduction

We seek pure traveling-wave profiles by introducing

$$\xi = \lambda_1 z + \lambda_2 t, \quad A(z, t) = A(\xi), \quad (2)$$

with λ_1, λ_2 real propagation constants. The chain rule gives

$$\frac{\partial A}{\partial z} = \lambda_1 \frac{dA}{d\xi}, \quad \frac{\partial^2 A}{\partial t^2} = \lambda_2^2 \frac{d^2 A}{d\xi^2}, \quad (3)$$

$$\frac{\partial^3 A}{\partial t^3} = \lambda_2^3 \frac{d^3 A}{d\xi^3}, \quad \frac{\partial^4 A}{\partial t^4} = \lambda_2^4 \frac{d^4 A}{d\xi^4}. \quad (4)$$

Substituting into (1) yields the traveling-wave ODE

$$\lambda_1 A' + \frac{i}{2}(\beta_2 + ig_0 T_2^2) \lambda_2^2 A'' + \frac{\beta_3}{6} \lambda_2^3 A''' + \frac{i\beta_4}{24} \lambda_2^4 A^{(4)} + i\left(\gamma + i\frac{\alpha_2}{2}\right) \frac{|A|^2 A}{1 + \Gamma|A|^2} + \frac{1}{2}(g_0 - \alpha)A = 0, \quad (5)$$

where primes denote $d/d\xi$.

4 The iB-Function (BDKm) Ansatz

We represent the traveling-wave profile with the implicit Bogning function (iB-function) ansatz,

$$A(\xi) = a J_{n,m}(\xi), \quad J_{n,m}(\xi) = \frac{\sinh^m(\xi)}{\cosh^n(\xi)}, \quad (6)$$

where a is a (generally complex) amplitude and n, m are exponents to be determined. The ansatz obeys the closed differentiation rule

$$\frac{d}{d\xi} J_{n,m} = m J_{n-1,m-1} - n J_{n+1,m+1}, \quad (7)$$

which follows directly from $\frac{d}{d\xi} \sinh = \cosh$, $\frac{d}{d\xi} \cosh = \sinh$ and the quotient rule. Iterating (7) gives the derivatives of $A(\xi)$ used below.

4.1 First and second derivatives

$$A' = a [m J_{n-1,m-1} - n J_{n+1,m+1}], \quad (8)$$

$$A'' = a [m(m-1) J_{n-2,m-2} - (m(n-1) + n(m+1)) J_{n,m} + n(n+1) J_{n+2,m+2}]. \quad (9)$$

4.2 Third derivative

$$A''' = a [A_1 J_{n-3,m-3} + A_2 J_{n-1,m-1} + A_3 J_{n+1,m+1} + A_4 J_{n+3,m+3}], \quad (10)$$

with

$$A_1 = m(m-1)(m-2) = m^3 - 3m^2 + 2m, \quad (11)$$

$$A_2 = -3m^2 n + 3m^2 - 2m, \quad (12)$$

$$A_3 = 3mn^2 + 3n^2 + 2n, \quad (13)$$

$$A_4 = -n(n+1)(n+2) = -n^3 - 3n^2 - 2n. \quad (14)$$

4.3 Fourth derivative

$$\begin{aligned}
A^{(4)} = a \Big\{ & A_1(m-3)J_{n-4,m-4} + [A_2(m-1) - A_1(n-3)]J_{n-2,m-2} \\
& + [A_3(m+1) - A_2(n-1)]J_{n,m} + [A_4(m+3) - A_3(n+1)]J_{n+2,m+2} \\
& - A_4(n+3)J_{n+4,m+4} \Big\}.
\end{aligned} \tag{15}$$

5 Substitution and Coefficient Grouping

Using the power identities $|A|^2 A = a^3 J_{3n,3m}$ and $1 + \Gamma|A|^2 = 1 + \Gamma a^2 J_{2n,2m}$ (taking a real without loss of generality for the modulus), substituting the derivatives above into (5) and dividing by a gives, before clearing the saturation denominator,

$$\left[\sum_{k=1}^9 C_k J_{n+r_k, m+r_k} \right] + i(\gamma + i\frac{\alpha_2}{2}) \frac{a^2 J_{3n,3m}}{1 + \Gamma a^2 J_{2n,2m}} = 0, \tag{16}$$

with offsets $r_k \in \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$ for $k = 1, \dots, 9$ and the grouped coefficients

$$C_1 = \frac{i\beta_4}{24} \lambda_2^4 m(m-1)(m-2)(m-3), \tag{17}$$

$$C_2 = \frac{\beta_3}{6} \lambda_2^3 m(m-1)(m-2), \tag{18}$$

$$C_3 = \frac{i}{2} (\beta_2 + ig_0 T_2^2) \lambda_2^2 m(m-1) + \frac{i\beta_4}{24} \lambda_2^4 [(-3m^2 n + 3m^2 - 2m)(m-1) - m(m-1)(m-2)(n-3)], \tag{19}$$

$$C_4 = \lambda_1 m + \frac{\beta_3}{6} \lambda_2^3 (-3m^2 n + 3m^2 - 2m), \tag{20}$$

$$\begin{aligned}
C_5 = & \frac{1}{2} (g_0 - \alpha) - \frac{i}{2} (\beta_2 + ig_0 T_2^2) \lambda_2^2 [m(n-1) + n(m+1)] \\
& + \frac{i\beta_4}{24} \lambda_2^4 [(3mn^2 + 3n^2 + 2n)(m+1) - (-3m^2 n + 3m^2 - 2m)(n-1)],
\end{aligned} \tag{21}$$

$$C_6 = -\lambda_1 n + \frac{\beta_3}{6} \lambda_2^3 (3mn^2 + 3n^2 + 2n), \tag{22}$$

$$C_7 = \frac{i}{2} (\beta_2 + ig_0 T_2^2) \lambda_2^2 n(n+1) + \frac{i\beta_4}{24} \lambda_2^4 [-n(n+1)(n+2)(m+3) - (3mn^2 + 3n^2 + 2n)(n+1)], \tag{23}$$

$$C_8 = -\frac{\beta_3}{6} \lambda_2^3 n(n+1)(n+2), \tag{24}$$

$$C_9 = \frac{i\beta_4}{24} \lambda_2^4 n(n+1)(n+2)(n+3). \tag{25}$$

6 The Fully Expanded Range Equation

Multiplying (16) by $1 + \Gamma a^2 J_{2n,2m}$ and using $J_{2n,2m} \cdot J_{p,q} = J_{2n+p,2m+q}$ term by term gives the complete, fraction-free 19-term *Range equation*:

$$\begin{aligned}
& C_1 J_{n-4,m-4} + C_2 J_{n-3,m-3} + C_3 J_{n-2,m-2} + C_4 J_{n-1,m-1} + C_5 J_{n,m} \\
& + C_6 J_{n+1,m+1} + C_7 J_{n+2,m+2} + C_8 J_{n+3,m+3} + C_9 J_{n+4,m+4} \\
& + \Gamma a^2 \left[C_1 J_{3n-4,3m-4} + C_2 J_{3n-3,3m-3} + C_3 J_{3n-2,3m-2} + C_4 J_{3n-1,3m-1} + C_5 J_{3n,3m} \right. \\
& \quad \left. + C_6 J_{3n+1,3m+1} + C_7 J_{3n+2,3m+2} + C_8 J_{3n+3,3m+3} + C_9 J_{3n+4,3m+4} \right] \\
& + i \left(\gamma + i \frac{\alpha_2}{2} \right) a^2 J_{3n,3m} = 0.
\end{aligned} \tag{26}$$

For a solution to exist, every coefficient of every distinct $J_{p,q}$ index appearing in (26) must vanish (or the terms with matching indices must sum to zero). Because the first block has indices $n+r$ ($r \in \{-4, \dots, 4\}$) and the second block has indices $3n+s$ ($s \in \{-4, \dots, 4\}$), a term from each block coincides only for special (n, m) ; this motivates the combinatorial classification below.

7 Combinatorial Classification of Admissible (n, m)

7.1 Candidate values

Let $r, s \in \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$ index a term $J_{n+r,m+r}$ in the first block and $J_{3n+s,3m+s}$ in the second block of (26). Equality of the first indices, $n+r = 3n+s$, requires

$$n = \frac{r-s}{2}, \tag{27}$$

and equality of the second indices gives the identical relation for m . Since a single J -function pair must match simultaneously in both indices for the same (r, s) ,

$$n = m = \frac{r-s}{2}. \tag{28}$$

Table 1 tabulates $n = (r-s)/2$ over all 81 ordered pairs (r, s) .

Table 1: Values of $n = (r-s)/2$ obtained from $n+r = 3n+s$, for $r, s \in \{-4, \dots, 4\}$.

$r \backslash s$	-4	-3	-2	-1	0	1	2	3	4
-4	0	$-\frac{1}{2}$	-1	$-\frac{3}{2}$	-2	$-\frac{5}{2}$	-3	$-\frac{7}{2}$	-4
-3	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{3}{2}$	-2	$-\frac{5}{2}$	-3	$-\frac{7}{2}$
-2	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{3}{2}$	-2	$-\frac{5}{2}$	-3
-1	$\frac{3}{2}$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{3}{2}$	-2	$-\frac{5}{2}$
0	2	$\frac{3}{2}$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{3}{2}$	-2
1	$\frac{5}{2}$	2	$\frac{3}{2}$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{3}{2}$
2	3	$\frac{5}{2}$	2	$\frac{3}{2}$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1
3	$\frac{7}{2}$	3	$\frac{5}{2}$	2	$\frac{3}{2}$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$
4	4	$\frac{7}{2}$	3	$\frac{5}{2}$	2	$\frac{3}{2}$	1	$\frac{1}{2}$	0

Although there are 81 ordered comparisons, they collapse to only 17 distinct candidate pairs $(n, m) = (q, q)$ with

$$q \in \left\{ -4, -\frac{7}{2}, -3, -\frac{5}{2}, -2, -\frac{3}{2}, -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, 3, \frac{7}{2}, 4 \right\}. \tag{29}$$

Restricting to integer (n, m) leaves $q = -4, \dots, 4$; further restricting to nonnegative integers leaves $q = 0, 1, 2, 3, 4$.

7.2 Combinatorial frequency of each candidate

Assigning each of the 81 ordered comparisons (r, s) equal weight $1/81$, the number of comparisons yielding a given difference $d = r - s = 2q$ is $N_d = 9 - |d|$, so

$$N(q) = 9 - 2|q|, \quad P(q, q) = \frac{N(q)}{81} = \frac{9 - 2|q|}{81}, \quad q = -4, -\frac{7}{2}, \dots, \frac{7}{2}, 4. \quad (30)$$

We stress that $P(q, q)$ is a purely combinatorial frequency over index comparisons, not a physical probability of solution existence; it is a convenient measure of how many index-matching routes support a given candidate pair. Table 2 lists the full distribution, which sums to unity by construction.

Table 2: Combinatorial frequency of each candidate pair $(n, m) = (q, q)$.

(n, m)	Appearances	Probability	Percentage
$(-4, -4)$	1	$1/81$	1.23%
$(-7/2, -7/2)$	2	$2/81$	2.47%
$(-3, -3)$	3	$3/81$	3.70%
$(-5/2, -5/2)$	4	$4/81$	4.94%
$(-2, -2)$	5	$5/81$	6.17%
$(-3/2, -3/2)$	6	$6/81$	7.41%
$(-1, -1)$	7	$7/81$	8.64%
$(-1/2, -1/2)$	8	$8/81$	9.88%
$(0, 0)$	9	$9/81$	11.11%
$(1/2, 1/2)$	8	$8/81$	9.88%
$(1, 1)$	7	$7/81$	8.64%
$(3/2, 3/2)$	6	$6/81$	7.41%
$(2, 2)$	5	$5/81$	6.17%
$(5/2, 5/2)$	4	$4/81$	4.94%
$(3, 3)$	3	$3/81$	3.70%
$(7/2, 7/2)$	2	$2/81$	2.47%
$(4, 4)$	1	$1/81$	1.23%

The most combinatorially supported candidate is $(0, 0)$, corresponding to the trivial constant $J_{0,0} \equiv 1$; the extreme pairs $(\pm 4, \pm 4)$ are supported by a single comparison route each.

7.3 The restricted field

The full extended field formed from the 17 candidate values in each coordinate contains $17^2 = 289$ pairs. We restrict attention to the field generated by the six most combinatorially supported nontrivial candidates together with $(0, 0)$,

$$\mathcal{F}_R = \left\{ -\frac{3}{2}, -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2} \right\}^2, \quad (31)$$

which contains 49 ordered pairs (n, m) . For every pair in $\mathcal{F}_R$ the coefficients $C_1, \dots, C_9$ can be evaluated by exact substitution of n, m into the expressions of Section 5. The full 49-branch table is given in Appendix A.

8 A Worked Branch: $(n, m) = (1, 1)$

To make the general procedure concrete, we work through the branch $(n, m) = (1, 1)$, for which $J_{1,1}(\xi) = \tanh(\xi)$, so the ansatz is the familiar dark/kink profile $A(\xi) = a \tanh(\xi)$. Evaluating the coefficient expressions of Section 5 at $n = m = 1$ gives (Appendix A, Table 14)

$$C_1 = 0, \quad C_2 = 0, \quad C_3 = 0, \quad (32)$$

$$C_4 = \lambda_1 - \frac{\beta_3}{3} \lambda_2^3, \quad C_5 = \frac{g_0 - \alpha}{2} - i \left(\beta_2 + i \frac{g_0 T_2^2}{2} \right) \lambda_2^2 + \frac{2i\beta_4}{3} \lambda_2^4, \quad (33)$$

$$C_6 = -\lambda_1 + \frac{4\beta_3}{3} \lambda_2^3, \quad C_7 = i \left(\beta_2 + i \frac{g_0 T_2^2}{2} \right) \lambda_2^2 - \frac{5i\beta_4}{3} \lambda_2^4, \quad (34)$$

$$C_8 = -\beta_3 \lambda_2^3, \quad C_9 = i\beta_4 \lambda_2^4. \quad (35)$$

The vanishing of C_1, C_2, C_3 is not a coincidence: for $n = m = 1$ the offset indices $n - 4, n - 3, n - 2$ correspond to $J_{-3,-3}, J_{-2,-2}, J_{-1,-1}$, and the prefactors $A_1(m - 3), A_2(m - 1) - A_1(n - 3)$, and the second-derivative bracket all contain a factor of $(m - 1)$ or $(m - 1)(m - 2) \cdots$ that vanishes identically at $m = 1$. This is a generic feature of integer branches with $n = m$: the “deep negative offset” terms of the fourth derivative collapse automatically, leaving a shorter, more tractable coefficient system. The remaining Range equation for this branch, after also imposing the matching of $J_{3,3} = \tanh^3(\xi)$ terms coming from the saturable-nonlinearity expansion and the self-phase-modulation term, is

$$C_4 J_{0,0} + C_5 J_{1,1} + C_6 J_{2,2} + C_7 J_{2,2} + \Gamma a^2 [\cdots]_{J_{3,3}, J_{5,5}, \dots} + i \left(\gamma + i \frac{\alpha_2}{2} \right) a^2 J_{3,3} = 0, \quad (36)$$

and setting the coefficient of each independent power of $\tanh(\xi)$ to zero yields an algebraic system relating a, λ_1 to $\beta_2, \beta_3, \beta_4, g_0, T_2, \alpha, \alpha_2, \gamma, \Gamma$ (most directly: $C_4 = 0$ fixes $\lambda_1 = \beta_3 \lambda_2^3 / 3$ once the cubic-dispersion branch is selected, and the remaining balance between $C_5 + C_6 + C_7$ and the nonlinear/saturation terms fixes a^2 in terms of the gain-loss and dispersion parameters). We do not carry every branch of $\mathcal{F}_R$ to this level of explicit closure in this paper; the point of Table 2–16 is to make the coefficient system for *any* of the 49 branches immediately available for such a reduction, whether performed by hand or automatically with a computer-algebra or symbolic-regression backend (Section 9).

9 Application: A Benchmark Design for Physics-Informed Neural Networks

9.1 Motivation

PINNs [1] approximate the field $A(z, t) = u(z, t) + iv(z, t)$ by a neural network $\hat{A}_\theta$ and train it by minimizing a composite loss built from (i) a data-fit term against known values (initial/boundary conditions, or sparse measurements) and (ii) a physics-residual term obtained by substituting $\hat{A}_\theta$ into the governing PDE and evaluating the residual on collocation points via automatic differentiation. This scheme has been applied successfully to the NLS equation, the Hirota (third-order NLS) equation [2], and higher-order NLS equations with self-steepening and Raman terms [3], in each case validated against one or a handful of hand-picked exact solutions with fixed coefficients. A structural limitation of this approach is that it cannot easily test *generalization*: whether a network (or a class of network architectures/training schedules) that performs well on one soliton also performs well across a structured family of solutions with different exponent structure and different physical-parameter regimes.

The restricted field $\mathcal{F}_R$ derived in Section 7 directly addresses this limitation for the higher-order gain-saturated equation (1): it provides 49 closed-form solution branches, each an exact solution of (1) once the associated algebraic constraint system (as illustrated in Section 8) is imposed on $(a, \lambda_1, \lambda_2, \beta_2, \beta_3, \beta_4, g_0, T_2, \alpha, \alpha_2, \gamma, \Gamma)$. Because the branches differ systematically in (n, m) – and hence in qualitative solution shape (bright, dark, singular, or mixed hyperbolic profiles) – they form a natural train/held-out split for generalization testing that a single soliton solution cannot provide.

9.2 Forward-problem loss functional

Writing $A = u + iv$ and the residual $F = F_u + iF_v$ obtained from (1),

$$F_u := u_z - \frac{1}{2}(\beta_2 v_{tt} + g_0 T_2^2 u_{tt}) - \frac{\beta_3}{6} v_{ttt} - \frac{\beta_4}{24} u_{tttt} - (\gamma + i\frac{\alpha_2}{2}) \Re \left[i \frac{(u^2 + v^2)(u + iv)}{1 + \Gamma(u^2 + v^2)} \right] + \frac{1}{2}(g_0 - \alpha)u, \quad (37)$$

$$F_v := v_z + \frac{1}{2}(\beta_2 u_{tt} - g_0 T_2^2 v_{tt}) + \frac{\beta_3}{6} u_{ttt} - \frac{\beta_4}{24} v_{tttt} - (\gamma + i\frac{\alpha_2}{2}) \Im \left[i \frac{(u^2 + v^2)(u + iv)}{1 + \Gamma(u^2 + v^2)} \right] + \frac{1}{2}(g_0 - \alpha)v, \quad (38)$$

the forward-problem training loss for a given branch $(n, m) \in \mathcal{F}_R$, with its exact solution $A_{n,m}(\xi; \text{params})$ supplying initial/boundary data, is

$$\begin{aligned} \mathcal{L}_{\text{fwd}} = & \frac{1}{N_I} \sum_{j=1}^{N_I} |\hat{A}_\theta(z_0, t_I^j) - A_{n,m}(z_0, t_I^j)|^2 + \frac{1}{N_B} \sum_{j=1}^{N_B} |\hat{A}_\theta(z, \pm L_j) - A_{n,m}(z, \pm L_j)|^2 \\ & + \frac{1}{N_S} \sum_{j=1}^{N_S} (F_u^2 + F_v^2) \Big|_{(z_S^j, t_S^j)}, \end{aligned} \quad (39)$$

following the standard Latin-hypercube collocation strategy used throughout this literature [1–3].

9.3 Inverse-problem (parameter-discovery) loss functional

For parameter discovery, $\beta_2, \beta_3, \beta_4, g_0, T_2, \alpha, \alpha_2, \gamma, \Gamma$ (or any chosen subset) are promoted to trainable scalars alongside the network weights, and the loss becomes

$$\mathcal{L}_{\text{inv}} = \frac{1}{N_p} \sum_{j=1}^{N_p} |\hat{A}_\theta(z^j, t^j) - A_{n,m}^j|^2 + \frac{1}{N_p} \sum_{j=1}^{N_p} (F_u^2 + F_v^2) \Big|_{(z^j, t^j)}, \quad (40)$$

with $\{z^j, t^j, A_{n,m}^j\}$ sampled from one exact branch. This is a direct generalization of the two- and four-parameter discovery experiments of Zhou and Yan [2] and Fang et al. [3] to a model with up to nine independent physical coefficients; the restricted field $\mathcal{F}_R$ makes it possible to test parameter recovery separately on bright-type branches (e.g. small $|n|, |m|$) and singular/kink-type branches (e.g. $n = m = 1$), which is not possible with a single soliton family.

9.4 Proposed generalization-testing protocol

We propose the following protocol as future numerical work built directly on the catalog in Appendix A:

- (i) **Branch generation.** For each $(n, m) \in \mathcal{F}_R$, solve the algebraic closure conditions illustrated in Section 8 (symbolically, e.g. with a computer-algebra system) to obtain the admissible parameter manifold and one or more representative parameter sets; sample $\{z^j, t^j, A^j\}$ from each resulting exact solution.
- (ii) **Held-out split.** Partition $\mathcal{F}_R$ into a training subset (e.g. $|n|, |m| \leq 1$) and a held-out subset (e.g. $|n|$ or $|m| = 3/2$), train a single PINN architecture on the training subset only, and evaluate the relative L^2 error on the held-out branches.
- (iii) **Parameter-recovery robustness.** Repeat the inverse-problem experiment of Zhou and Yan [2], Fang et al. [3] independently on each branch and report how recovery error for $(\beta_2, \beta_3, \beta_4, g_0, T_2, \alpha, \alpha_2, \gamma, \Gamma)$ varies with branch type, under both clean and noise-perturbed data (as in Zhou and Yan [2]).
- (iv) **Ablation on saturation.** Because Γ enters only through the denominator $1 + \Gamma|A|^2$, compare PINN performance with $\Gamma = 0$ (unsaturated, standard higher-order NLS) against $\Gamma \neq 0$ branches, to quantify whether the saturable nonlinearity increases the forward- or inverse-problem difficulty.

This protocol, and the underlying 49-branch catalog, is the concrete deliverable this paper contributes to future PINN work on higher-order gain-saturated evolution equations; we emphasize again that we report the framework and worked derivation only, not executed numerical experiments.

10 Discussion

The combinatorial classification of Section 7 is, to our knowledge, not typically made explicit in the ansatz-method literature, where authors usually state the admissible (n, m) (or equivalent tanh/sech exponents) without enumerating the full comparison space or reporting a frequency measure over it. Making this step explicit and reproducible is what allows the resulting solution catalog to be generated systematically (Appendix A) rather than one branch at a time, which is what makes it usable as a benchmark generator rather than a single validation example. Existing symbolic-computation packages for the tanh/sech method [6] automate a closely related search over exponents for a narrower ansatz family; extending that kind of automation to the iB-function family $J_{n,m}$ with fractional exponents, and coupling it directly to PINN-benchmark generation as specified in Section 9, is a natural direction for follow-up software work.

We also note two limitations. First, not every branch in $\mathcal{F}_R$ will admit a non-degenerate closure (some coefficient systems may be over- or under-determined, forcing $a = 0$ or leaving free parameters unconstrained); a systematic audit of which of the 49 branches yield genuine, non-trivial exact solutions is needed before the full catalog can be used as-is as a benchmark suite. Second, the combinatorial frequency $P(q, q)$ of Eq. (30) measures index-matching multiplicity, not solution existence or physical relevance, and should not be over-interpreted as a likelihood that a given branch yields a physically meaningful pulse.

11 Conclusion

We have derived the traveling-wave reduction of a fourth-order gain-saturated nonlinear Schrödinger-type equation under the iB-function (BDKm) ansatz, obtained the fully expanded Range equation, and carried out a complete combinatorial classification of the admissible index pairs (n, m) , yielding

a restricted field of 49 branches with fully expanded closed-form coefficients in terms of all eight physical parameters of the model. We have shown, through the explicit $(n, m) = (1, 1)$ example, how each branch reduces to an algebraic closure system fixing the wave parameters in terms of the physical ones. Building on this catalog, we specified a concrete benchmark design for physics-informed neural networks – forward-problem and inverse-problem loss functionals matching the full nine-parameter model, and a generalization-testing protocol exploiting the branch structure of $\mathcal{F}_R$ – that addresses a specific limitation of existing PINN-for-NLSE-hierarchy studies, namely their reliance on isolated, hand-picked solutions rather than structured families. Executing this protocol numerically, and auditing the non-degeneracy of all 49 branches, are the natural next steps.

A Fully Expanded Coefficient Tables for the Restricted Field

For every $(n, m) \in \mathcal{F}_R = \{-3/2, -1, -1/2, 0, 1/2, 1, 3/2\}^2$ we report the exact, simplified values of $C_1, \dots, C_9$ from Section 5, grouped by the index n .

A.1 $n = -3/2$

Table 3: Coefficients for $n = -3/2$, $m \in \{-3/2, -1, -1/2\}$.

Coeff.	$(-3/2, -3/2)$	$(-3/2, -1)$	$(-3/2, -1/2)$	
C_1	$\frac{315i\beta_4}{128}\lambda_2^4$	$i\beta_4\lambda_2^4$	$\frac{35i\beta_4}{128}\lambda_2^4$	$\Xi := \beta_2 + i g_0 T_2^2$
C_2	$-\frac{35\beta_3}{16}\lambda_2^3$	$-\beta_3\lambda_2^3$	$-\frac{5\beta_3}{16}\lambda_2^3$	
C_3	$\frac{15i}{8}\Xi\lambda_2^2 - \frac{145i\beta_4}{32}\lambda_2^4$	$i\Xi\lambda_2^2 - \frac{23i\beta_4}{12}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{17i\beta_4}{32}\lambda_2^4$	
C_4	$-\frac{3}{2}\lambda_1 + \frac{53\beta_3}{16}\lambda_2^3$	$-\lambda_1 + \frac{19\beta_3}{12}\lambda_2^3$	$-\frac{1}{2}\lambda_1 + \frac{23\beta_3}{48}\lambda_2^3$	
C_5	$\frac{g_0 - \alpha}{2} - \frac{9i}{4}\Xi\lambda_2^2 + \frac{141i\beta_4}{64}\lambda_2^4$	$\frac{g_0 - \alpha}{2} - \frac{5i}{4}\Xi\lambda_2^2 + \frac{95i\beta_4}{96}\lambda_2^4$	$\frac{g_0 - \alpha}{2} - \frac{i}{4}\Xi\lambda_2^2 + \frac{59i\beta_4}{192}\lambda_2^4$	
C_6	$\frac{3}{2}\lambda_1 - \frac{17\beta_3}{16}\lambda_2^3$	$\frac{3}{2}\lambda_1 - \frac{\beta_3}{2}\lambda_2^3$	$\frac{3}{2}\lambda_1 + \frac{\beta_3}{16}\lambda_2^3$	
C_7	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{5i\beta_4}{32}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{3i\beta_4}{32}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{i\beta_4}{32}\lambda_2^4$	
C_8	$-\frac{\beta_3}{16}\lambda_2^3$	$-\frac{\beta_3}{16}\lambda_2^3$	$-\frac{\beta_3}{16}\lambda_2^3$	
C_9	$\frac{3i\beta_4}{128}\lambda_2^4$	$\frac{3i\beta_4}{128}\lambda_2^4$	$\frac{3i\beta_4}{128}\lambda_2^4$	

throughout the appendix.

Table 4: Coefficients for $n = -3/2$, $m \in \{0, 1/2, 1, 3/2\}$.

Coeff.	$(-3/2, 0)$	$(-3/2, 1/2)$	$(-3/2, 1)$	$(-3/2, 3/2)$
C_1	0	$-\frac{5i\beta_4}{128}\lambda_2^4$	0	$\frac{3i\beta_4}{128}\lambda_2^4$
C_2	0	$\frac{\beta_3}{16}\lambda_2^3$	0	$-\frac{\beta_3}{16}\lambda_2^3$
C_3	0	$-\frac{i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{96}\lambda_2^4$	0	$\frac{3i}{8}\Xi\lambda_2^2 + \frac{7i\beta_4}{32}\lambda_2^4$
C_4	0	$\frac{1}{2}\lambda_1 + \frac{7\beta_3}{48}\lambda_2^3$	$\lambda_1 + \frac{11\beta_3}{12}\lambda_2^3$	$\frac{3}{2}\lambda_1 + \frac{37\beta_3}{16}\lambda_2^3$
C_5	$\frac{g_0 - \alpha}{2} + \frac{3i}{4}\Xi\lambda_2^2 + \frac{5i\beta_4}{32}\lambda_2^4$	$\frac{g_0 - \alpha}{2} + \frac{7i}{4}\Xi\lambda_2^2 + \frac{103i\beta_4}{192}\lambda_2^4$	$\frac{g_0 - \alpha}{2} + \frac{11i}{4}\Xi\lambda_2^2 + \frac{139i\beta_4}{96}\lambda_2^4$	$\frac{g_0 - \alpha}{2} + \frac{15i}{4}\Xi\lambda_2^2 + \frac{185i\beta_4}{64}\lambda_2^4$
C_6	$\frac{3}{2}\lambda_1 + \frac{5\beta_3}{8}\lambda_2^3$	$\frac{3}{2}\lambda_1 + \frac{19\beta_3}{16}\lambda_2^3$	$\frac{3}{2}\lambda_1 + \frac{7\beta_3}{4}\lambda_2^3$	$\frac{3}{2}\lambda_1 + \frac{37\beta_3}{16}\lambda_2^3$
C_7	$\frac{3i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{32}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 + \frac{3i\beta_4}{32}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{32}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 + \frac{7i\beta_4}{32}\lambda_2^4$
C_8	$-\frac{\beta_3}{16}\lambda_2^3$	$-\frac{\beta_3}{16}\lambda_2^3$	$-\frac{\beta_3}{16}\lambda_2^3$	$-\frac{\beta_3}{16}\lambda_2^3$
C_9	$\frac{3i\beta_4}{128}\lambda_2^4$	$\frac{3i\beta_4}{128}\lambda_2^4$	$\frac{3i\beta_4}{128}\lambda_2^4$	$\frac{3i\beta_4}{128}\lambda_2^4$

Table 5: Coefficients for $n = -1$, $m \in \{-3/2, -1, -1/2\}$.

Coeff.	$(-1, -3/2)$	$(-1, -1)$	$(-1, -1/2)$
C_1	$\frac{315i\beta_4}{128}\lambda_2^4$	$i\beta_4\lambda_2^4$	$\frac{35i\beta_4}{128}\lambda_2^4$
C_2	$-\frac{35\beta_3}{16}\lambda_2^3$	$-\beta_3\lambda_2^3$	$-\frac{5\beta_3}{16}\lambda_2^3$
C_3	$\frac{15i}{8}\Xi\lambda_2^2 - \frac{125i\beta_4}{32}\lambda_2^4$	$i\Xi\lambda_2^2 - \frac{5i\beta_4}{3}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{15i\beta_4}{32}\lambda_2^4$
C_4	$-\frac{3}{2}\lambda_1 + \frac{11\beta_3}{4}\lambda_2^3$	$-\lambda_1 + \frac{4\beta_3}{3}\lambda_2^3$	$-\frac{1}{2}\lambda_1 + \frac{5\beta_3}{12}\lambda_2^3$
C_5	$\frac{g_0-\alpha}{2} - \frac{7i}{4}\Xi\lambda_2^2 + \frac{139i\beta_4}{96}\lambda_2^4$	$\frac{g_0-\alpha}{2} - i\Xi\lambda_2^2 + \frac{2i\beta_4}{3}\lambda_2^4$	$\frac{g_0-\alpha}{2} - \frac{i}{4}\Xi\lambda_2^2 + \frac{19i\beta_4}{96}\lambda_2^4$
C_6	$\lambda_1 - \frac{7\beta_3}{12}\lambda_2^3$	$\lambda_1 - \frac{\beta_3}{3}\lambda_2^3$	$\lambda_1 - \frac{\beta_3}{12}\lambda_2^3$
C_7	0	0	0
C_8	0	0	0
C_9	0	0	0

 Table 6: Coefficients for $n = -1$, $m \in \{0, 1/2, 1, 3/2\}$.

Coeff.	$(-1, 0)$	$(-1, 1/2)$	$(-1, 1)$	$(-1, 3/2)$
C_1	0	$-\frac{5i\beta_4}{128}\lambda_2^4$	0	$\frac{3i\beta_4}{128}\lambda_2^4$
C_2	0	$\frac{\beta_3}{16}\lambda_2^3$	0	$-\frac{\beta_3}{16}\lambda_2^3$
C_3	0	$-\frac{i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{96}\lambda_2^4$	0	$\frac{3i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{32}\lambda_2^4$
C_4	0	$\frac{1}{2}\lambda_1 + \frac{\beta_3}{12}\lambda_2^3$	$\lambda_1 + \frac{2\beta_3}{3}\lambda_2^3$	$\frac{3}{2}\lambda_1 + \frac{7\beta_3}{4}\lambda_2^3$
C_5	$\frac{g_0-\alpha}{2} + \frac{i}{2}\Xi\lambda_2^2 + \frac{i\beta_4}{24}\lambda_2^4$	$\frac{g_0-\alpha}{2} + \frac{5i}{4}\Xi\lambda_2^2 + \frac{19i\beta_4}{96}\lambda_2^4$	$\frac{g_0-\alpha}{2} + 2i\Xi\lambda_2^2 + \frac{2i\beta_4}{3}\lambda_2^4$	$\frac{g_0-\alpha}{2} + \frac{11i}{4}\Xi\lambda_2^2 + \frac{139i\beta_4}{96}\lambda_2^4$
C_6	$\lambda_1 + \frac{\beta_3}{6}\lambda_2^3$	$\lambda_1 + \frac{5\beta_3}{12}\lambda_2^3$	$\lambda_1 + \frac{2\beta_3}{3}\lambda_2^3$	$\lambda_1 + \frac{11\beta_3}{12}\lambda_2^3$
C_7	0	0	0	0
C_8	0	0	0	0
C_9	0	0	0	0

A.2 $n = -1$

A.3 $n = -1/2$

Table 7: Coefficients for $n = -1/2$, $m \in \{-3/2, -1, -1/2\}$.

Coeff.	$(-1/2, -3/2)$	$(-1/2, -1)$	$(-1/2, -1/2)$
C_1	$\frac{315i\beta_4}{128}\lambda_2^4$	$i\beta_4\lambda_2^4$	$\frac{35i\beta_4}{128}\lambda_2^4$
C_2	$-\frac{35\beta_3}{16}\lambda_2^3$	$-\beta_3\lambda_2^3$	$-\frac{5\beta_3}{16}\lambda_2^3$
C_3	$\frac{15i}{8}\Xi\lambda_2^2 - \frac{105i\beta_4}{32}\lambda_2^4$	$i\Xi\lambda_2^2 - \frac{17i\beta_4}{12}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{13i\beta_4}{32}\lambda_2^4$
C_4	$-\frac{3}{2}\lambda_1 + \frac{35\beta_3}{16}\lambda_2^3$	$-\lambda_1 + \frac{13\beta_3}{12}\lambda_2^3$	$-\frac{1}{2}\lambda_1 + \frac{17\beta_3}{48}\lambda_2^3$
C_5	$\frac{g_0-\alpha}{2} - \frac{5i}{4}\Xi\lambda_2^2 + \frac{163i\beta_4}{192}\lambda_2^4$	$\frac{g_0-\alpha}{2} - \frac{3i}{4}\Xi\lambda_2^2 + \frac{13i\beta_4}{32}\lambda_2^4$	$\frac{g_0-\alpha}{2} - \frac{i}{4}\Xi\lambda_2^2 + \frac{23i\beta_4}{192}\lambda_2^4$
C_6	$\frac{1}{2}\lambda_1 - \frac{11\beta_3}{48}\lambda_2^3$	$\frac{1}{2}\lambda_1 - \frac{\beta_3}{6}\lambda_2^3$	$\frac{1}{2}\lambda_1 - \frac{5\beta_3}{48}\lambda_2^3$
C_7	$-\frac{i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{96}\lambda_2^4$	$-\frac{i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{96}\lambda_2^4$	$-\frac{i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{96}\lambda_2^4$
C_8	$\frac{\beta_3}{16}\lambda_2^3$	$\frac{\beta_3}{16}\lambda_2^3$	$\frac{\beta_3}{16}\lambda_2^3$
C_9	$-\frac{5i\beta_4}{128}\lambda_2^4$	$-\frac{5i\beta_4}{128}\lambda_2^4$	$-\frac{5i\beta_4}{128}\lambda_2^4$

Table 8: Coefficients for $n = -1/2$, $m \in \{0, 1/2, 1, 3/2\}$.

Coeff.	$(-1/2, 0)$	$(-1/2, 1/2)$	$(-1/2, 1)$	$(-1/2, 3/2)$
C_1	0	$-\frac{5i\beta_4}{128}\lambda_2^4$	0	$\frac{3i\beta_4}{128}\lambda_2^4$
C_2	0	$\frac{\beta_3}{16}\lambda_2^3$	0	$-\frac{\beta_3}{16}\lambda_2^3$
C_3	0	$-\frac{i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{96}\lambda_2^4$	0	$\frac{3i}{8}\Xi\lambda_2^2 + \frac{3i\beta_4}{32}\lambda_2^4$
C_4	0	$\frac{1}{2}\lambda_1 + \frac{\beta_3}{48}\lambda_2^3$	$\lambda_1 + \frac{5\beta_3}{12}\lambda_2^3$	$\frac{3}{2}\lambda_1 + \frac{19\beta_3}{16}\lambda_2^3$
C_5	$\frac{g_0-\alpha}{2} + \frac{i}{4}\Xi\lambda_2^2 - \frac{i\beta_4}{96}\lambda_2^4$	$\frac{g_0-\alpha}{2} + \frac{3i}{4}\Xi\lambda_2^2 + \frac{i\beta_4}{64}\lambda_2^4$	$\frac{g_0-\alpha}{2} + \frac{5i}{4}\Xi\lambda_2^2 + \frac{19i\beta_4}{96}\lambda_2^4$	$\frac{g_0-\alpha}{2} + \frac{7i}{4}\Xi\lambda_2^2 + \frac{103i\beta_4}{192}\lambda_2^4$
C_6	$\frac{1}{2}\lambda_1 - \frac{\beta_3}{24}\lambda_2^3$	$\frac{1}{2}\lambda_1 + \frac{\beta_3}{48}\lambda_2^3$	$\frac{1}{2}\lambda_1 + \frac{\beta_3}{12}\lambda_2^3$	$\frac{1}{2}\lambda_1 + \frac{7\beta_3}{48}\lambda_2^3$
C_7	$-\frac{i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{96}\lambda_2^4$	$-\frac{i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{96}\lambda_2^4$	$-\frac{i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{96}\lambda_2^4$	$-\frac{i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{96}\lambda_2^4$
C_8	$\frac{\beta_3}{16}\lambda_2^3$	$\frac{\beta_3}{16}\lambda_2^3$	$\frac{\beta_3}{16}\lambda_2^3$	$\frac{\beta_3}{16}\lambda_2^3$
C_9	$-\frac{5i\beta_4}{128}\lambda_2^4$	$-\frac{5i\beta_4}{128}\lambda_2^4$	$-\frac{5i\beta_4}{128}\lambda_2^4$	$-\frac{5i\beta_4}{128}\lambda_2^4$

A.4 $n = 0$

A.5 $n = 1/2$

A.6 $n = 1$

A.7 $n = 3/2$

Data and Code Availability

The coefficient expressions in Appendix A were obtained by direct exact substitution of (n, m) into the closed-form expressions of Section 5 and are fully reproducible from that section alone. No numerical PINN experiments were executed for this paper; Section 9 specifies the methodology for future numerical work.

Table 9: Coefficients for $n = 0, m \in \{-3/2, -1, -1/2\}$.

Coeff.	$(0, -3/2)$	$(0, -1)$	$(0, -1/2)$
C_1	$\frac{315i\beta_4}{128}\lambda_2^4$	$i\beta_4\lambda_2^4$	$\frac{35i\beta_4}{128}\lambda_2^4$
C_2	$-\frac{35\beta_3}{16}\lambda_2^3$	$-\beta_3\lambda_2^3$	$-\frac{5\beta_3}{16}\lambda_2^3$
C_3	$\frac{15i}{8}\Xi\lambda_2^2 - \frac{85i\beta_4}{32}\lambda_2^4$	$i\Xi\lambda_2^2 - \frac{7i\beta_4}{6}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{11i\beta_4}{32}\lambda_2^4$
C_4	$-\frac{3}{2}\lambda_1 + \frac{13\beta_3}{8}\lambda_2^3$	$-\lambda_1 + \frac{5\beta_3}{6}\lambda_2^3$	$-\frac{1}{2}\lambda_1 + \frac{7\beta_3}{24}\lambda_2^3$
C_5	$\frac{g_0-\alpha}{2} - \frac{3i}{4}\Xi\lambda_2^2 + \frac{13i\beta_4}{32}\lambda_2^4$	$\frac{g_0-\alpha}{2} - \frac{i}{2}\Xi\lambda_2^2 + \frac{5i\beta_4}{24}\lambda_2^4$	$\frac{g_0-\alpha}{2} - \frac{i}{4}\Xi\lambda_2^2 + \frac{7i\beta_4}{96}\lambda_2^4$
C_6	0	0	0
C_7	0	0	0
C_8	0	0	0
C_9	0	0	0

Table 10: Coefficients for $n = 0, m \in \{0, 1/2, 1, 3/2\}$.

Coeff.	$(0, 0)$	$(0, 1/2)$	$(0, 1)$	$(0, 3/2)$
C_1	0	$-\frac{5i\beta_4}{128}\lambda_2^4$	0	$\frac{3i\beta_4}{128}\lambda_2^4$
C_2	0	$\frac{\beta_3}{16}\lambda_2^3$	0	$-\frac{\beta_3}{16}\lambda_2^3$
C_3	0	$-\frac{i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{96}\lambda_2^4$	0	$\frac{3i}{8}\Xi\lambda_2^2 + \frac{i\beta_4}{32}\lambda_2^4$
C_4	0	$\frac{1}{2}\lambda_1 - \frac{\beta_3}{24}\lambda_2^3$	$\lambda_1 + \frac{\beta_3}{6}\lambda_2^3$	$\frac{3}{2}\lambda_1 + \frac{5\beta_3}{8}\lambda_2^3$
C_5	$\frac{g_0-\alpha}{2}$	$\frac{g_0-\alpha}{2} + \frac{i}{4}\Xi\lambda_2^2 - \frac{i\beta_4}{96}\lambda_2^4$	$\frac{g_0-\alpha}{2} + \frac{i}{2}\Xi\lambda_2^2 + \frac{i\beta_4}{24}\lambda_2^4$	$\frac{g_0-\alpha}{2} + \frac{3i}{4}\Xi\lambda_2^2 + \frac{5i\beta_4}{32}\lambda_2^4$
C_6	0	0	0	0
C_7	0	0	0	0
C_8	0	0	0	0
C_9	0	0	0	0

Table 11: Coefficients for $n = 1/2, m \in \{-3/2, -1, -1/2\}$.

Coeff.	$(1/2, -3/2)$	$(1/2, -1)$	$(1/2, -1/2)$
C_1	$\frac{315i\beta_4}{128}\lambda_2^4$	$i\beta_4\lambda_2^4$	$\frac{35i\beta_4}{128}\lambda_2^4$
C_2	$-\frac{35\beta_3}{16}\lambda_2^3$	$-\beta_3\lambda_2^3$	$-\frac{5\beta_3}{16}\lambda_2^3$
C_3	$\frac{15i}{8}\Xi\lambda_2^2 - \frac{65i\beta_4}{32}\lambda_2^4$	$i\Xi\lambda_2^2 - \frac{11i\beta_4}{12}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{9i\beta_4}{32}\lambda_2^4$
C_4	$-\frac{3}{2}\lambda_1 + \frac{17\beta_3}{16}\lambda_2^3$	$-\lambda_1 + \frac{7\beta_3}{12}\lambda_2^3$	$-\frac{1}{2}\lambda_1 + \frac{11\beta_3}{48}\lambda_2^3$
C_5	$\frac{g_0-\alpha}{2} - \frac{i}{4}\Xi\lambda_2^2 + \frac{23i\beta_4}{192}\lambda_2^4$	$\frac{g_0-\alpha}{2} - \frac{i}{4}\Xi\lambda_2^2 + \frac{7i\beta_4}{96}\lambda_2^4$	$\frac{g_0-\alpha}{2} - \frac{i}{4}\Xi\lambda_2^2 + \frac{11i\beta_4}{192}\lambda_2^4$
C_6	$-\frac{1}{2}\lambda_1 + \frac{5\beta_3}{48}\lambda_2^3$	$-\frac{1}{2}\lambda_1 + \frac{\beta_3}{6}\lambda_2^3$	$-\frac{1}{2}\lambda_1 + \frac{11\beta_3}{48}\lambda_2^3$
C_7	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{5i\beta_4}{32}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{7i\beta_4}{32}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{9i\beta_4}{32}\lambda_2^4$
C_8	$-\frac{5\beta_3}{16}\lambda_2^3$	$-\frac{5\beta_3}{16}\lambda_2^3$	$-\frac{5\beta_3}{16}\lambda_2^3$
C_9	$\frac{35i\beta_4}{128}\lambda_2^4$	$\frac{35i\beta_4}{128}\lambda_2^4$	$\frac{35i\beta_4}{128}\lambda_2^4$

Table 12: Coefficients for $n = 1/2$, $m \in \{0, 1/2, 1, 3/2\}$.

Coeff.	(1/2, 0)	(1/2, 1/2)	(1/2, 1)	(1/2, 3/2)
C_1	0	$-\frac{5i\beta_4}{128}\lambda_2^4$	0	$\frac{3i\beta_4}{128}\lambda_2^4$
C_2	0	$\frac{\beta_3}{16}\lambda_2^3$	0	$-\frac{\beta_3}{16}\lambda_2^3$
C_3	0	$-\frac{i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{96}\lambda_2^4$	0	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{i\beta_4}{32}\lambda_2^4$
C_4	0	$\frac{1}{2}\lambda_1 - \frac{5\beta_3}{48}\lambda_2^3$	$\lambda_1 - \frac{\beta_3}{12}\lambda_2^3$	$\frac{3}{2}\lambda_1 + \frac{\beta_3}{16}\lambda_2^3$
C_5	$\frac{g_0-\alpha}{2} - \frac{i}{4}\Xi\lambda_2^2 + \frac{7i\beta_4}{96}\lambda_2^4$	$\frac{g_0-\alpha}{2} - \frac{i}{4}\Xi\lambda_2^2 + \frac{23i\beta_4}{192}\lambda_2^4$	$\frac{g_0-\alpha}{2} - \frac{i}{4}\Xi\lambda_2^2 + \frac{19i\beta_4}{96}\lambda_2^4$	$\frac{g_0-\alpha}{2} - \frac{i}{4}\Xi\lambda_2^2 + \frac{59i\beta_4}{192}\lambda_2^4$
C_6	$-\frac{1}{2}\lambda_1 + \frac{7\beta_3}{24}\lambda_2^3$	$-\frac{1}{2}\lambda_1 + \frac{17\beta_3}{48}\lambda_2^3$	$-\frac{1}{2}\lambda_1 + \frac{5\beta_3}{12}\lambda_2^3$	$-\frac{1}{2}\lambda_1 + \frac{23\beta_3}{48}\lambda_2^3$
C_7	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{11i\beta_4}{32}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{13i\beta_4}{32}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{15i\beta_4}{32}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{17i\beta_4}{32}\lambda_2^4$
C_8	$-\frac{5\beta_3}{16}\lambda_2^3$	$-\frac{5\beta_3}{16}\lambda_2^3$	$-\frac{5\beta_3}{16}\lambda_2^3$	$-\frac{5\beta_3}{16}\lambda_2^3$
C_9	$\frac{35i\beta_4}{128}\lambda_2^4$	$\frac{35i\beta_4}{128}\lambda_2^4$	$\frac{35i\beta_4}{128}\lambda_2^4$	$\frac{35i\beta_4}{128}\lambda_2^4$

Table 13: Coefficients for $n = 1$, $m \in \{-3/2, -1, -1/2\}$.

Coeff.	(1, -3/2)	(1, -1)	(1, -1/2)
C_1	$\frac{315i\beta_4}{128}\lambda_2^4$	$i\beta_4\lambda_2^4$	$\frac{35i\beta_4}{128}\lambda_2^4$
C_2	$-\frac{35\beta_3}{16}\lambda_2^3$	$-\beta_3\lambda_2^3$	$-\frac{5\beta_3}{16}\lambda_2^3$
C_3	$\frac{15i}{8}\Xi\lambda_2^2 - \frac{45i\beta_4}{32}\lambda_2^4$	$i\Xi\lambda_2^2 - \frac{2i\beta_4}{3}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{7i\beta_4}{32}\lambda_2^4$
C_4	$-\frac{3}{2}\lambda_1 + \frac{\beta_3}{2}\lambda_2^3$	$-\lambda_1 + \frac{\beta_3}{3}\lambda_2^3$	$-\frac{1}{2}\lambda_1 + \frac{\beta_3}{6}\lambda_2^3$
C_5	$\frac{g_0-\alpha}{2} + \frac{i}{4}\Xi\lambda_2^2 - \frac{i\beta_4}{96}\lambda_2^4$	$\frac{g_0-\alpha}{2}$	$\frac{g_0-\alpha}{2} - \frac{i}{4}\Xi\lambda_2^2 + \frac{7i\beta_4}{96}\lambda_2^4$
C_6	$-\lambda_1 + \frac{\beta_3}{12}\lambda_2^3$	$-\lambda_1 + \frac{\beta_3}{3}\lambda_2^3$	$-\lambda_1 + \frac{7\beta_3}{12}\lambda_2^3$
C_7	$i\Xi\lambda_2^2 - \frac{5i\beta_4}{12}\lambda_2^4$	$i\Xi\lambda_2^2 - \frac{2i\beta_4}{3}\lambda_2^4$	$i\Xi\lambda_2^2 - \frac{11i\beta_4}{12}\lambda_2^4$
C_8	$-\beta_3\lambda_2^3$	$-\beta_3\lambda_2^3$	$-\beta_3\lambda_2^3$
C_9	$i\beta_4\lambda_2^4$	$i\beta_4\lambda_2^4$	$i\beta_4\lambda_2^4$

Table 14: Coefficients for $n = 1$, $m \in \{0, 1/2, 1, 3/2\}$.

Coeff.	(1, 0)	(1, 1/2)	(1, 1)	(1, 3/2)
C_1	0	$-\frac{5i\beta_4}{128}\lambda_2^4$	0	$\frac{3i\beta_4}{128}\lambda_2^4$
C_2	0	$\frac{\beta_3}{16}\lambda_2^3$	0	$-\frac{\beta_3}{16}\lambda_2^3$
C_3	0	$-\frac{i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{96}\lambda_2^4$	0	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{3i\beta_4}{32}\lambda_2^4$
C_4	0	$\frac{1}{2}\lambda_1 - \frac{\beta_3}{6}\lambda_2^3$	$\lambda_1 - \frac{\beta_3}{3}\lambda_2^3$	$\frac{3}{2}\lambda_1 - \frac{\beta_3}{2}\lambda_2^3$
C_5	$\frac{g_0-\alpha}{2} - \frac{i}{2}\Xi\lambda_2^2 + \frac{5i\beta_4}{24}\lambda_2^4$	$\frac{g_0-\alpha}{2} - \frac{3i}{4}\Xi\lambda_2^2 + \frac{13i\beta_4}{32}\lambda_2^4$	$\frac{g_0-\alpha}{2} - i\Xi\lambda_2^2 + \frac{2i\beta_4}{3}\lambda_2^4$	$\frac{g_0-\alpha}{2} - \frac{5i}{4}\Xi\lambda_2^2 + \frac{95i\beta_4}{96}\lambda_2^4$
C_6	$-\lambda_1 + \frac{5\beta_3}{6}\lambda_2^3$	$-\lambda_1 + \frac{13\beta_3}{12}\lambda_2^3$	$-\lambda_1 + \frac{4\beta_3}{3}\lambda_2^3$	$-\lambda_1 + \frac{19\beta_3}{12}\lambda_2^3$
C_7	$i\Xi\lambda_2^2 - \frac{7i\beta_4}{6}\lambda_2^4$	$i\Xi\lambda_2^2 - \frac{17i\beta_4}{12}\lambda_2^4$	$i\Xi\lambda_2^2 - \frac{5i\beta_4}{3}\lambda_2^4$	$i\Xi\lambda_2^2 - \frac{23i\beta_4}{12}\lambda_2^4$
C_8	$-\beta_3\lambda_2^3$	$-\beta_3\lambda_2^3$	$-\beta_3\lambda_2^3$	$-\beta_3\lambda_2^3$
C_9	$i\beta_4\lambda_2^4$	$i\beta_4\lambda_2^4$	$i\beta_4\lambda_2^4$	$i\beta_4\lambda_2^4$

Table 15: Coefficients for $n = 3/2$, $m \in \{-3/2, -1, -1/2\}$.

Coeff.	$(3/2, -3/2)$	$(3/2, -1)$	$(3/2, -1/2)$
C_1	$\frac{315i\beta_4}{128}\lambda_2^4$	$i\beta_4\lambda_2^4$	$\frac{35i\beta_4}{128}\lambda_2^4$
C_2	$-\frac{35\beta_3}{16}\lambda_2^3$	$-\beta_3\lambda_2^3$	$-\frac{5\beta_3}{16}\lambda_2^3$
C_3	$\frac{15i}{8}\Xi\lambda_2^2 - \frac{25i\beta_4}{32}\lambda_2^4$	$i\Xi\lambda_2^2 - \frac{5i\beta_4}{12}\lambda_2^4$	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{5i\beta_4}{32}\lambda_2^4$
C_4	$-\frac{3}{2}\lambda_1 - \frac{\beta_3}{16}\lambda_2^3$	$-\lambda_1 + \frac{\beta_3}{12}\lambda_2^3$	$-\frac{1}{2}\lambda_1 + \frac{5\beta_3}{48}\lambda_2^3$
C_5	$\frac{g_0-\alpha}{2} + \frac{3i}{4}\Xi\lambda_2^2 + \frac{i\beta_4}{64}\lambda_2^4$	$\frac{g_0-\alpha}{2} + \frac{i}{4}\Xi\lambda_2^2 - \frac{i\beta_4}{96}\lambda_2^4$	$\frac{g_0-\alpha}{2} - \frac{i}{4}\Xi\lambda_2^2 + \frac{23i\beta_4}{192}\lambda_2^4$
C_6	$-\frac{3}{2}\lambda_1 - \frac{\beta_3}{16}\lambda_2^3$	$-\frac{3}{2}\lambda_1 + \frac{\beta_3}{2}\lambda_2^3$	$-\frac{3}{2}\lambda_1 + \frac{17\beta_3}{16}\lambda_2^3$
C_7	$\frac{15i}{8}\Xi\lambda_2^2 - \frac{25i\beta_4}{32}\lambda_2^4$	$\frac{15i}{8}\Xi\lambda_2^2 - \frac{45i\beta_4}{32}\lambda_2^4$	$\frac{15i}{8}\Xi\lambda_2^2 - \frac{65i\beta_4}{32}\lambda_2^4$
C_8	$-\frac{35\beta_3}{16}\lambda_2^3$	$-\frac{35\beta_3}{16}\lambda_2^3$	$-\frac{35\beta_3}{16}\lambda_2^3$
C_9	$\frac{315i\beta_4}{128}\lambda_2^4$	$\frac{315i\beta_4}{128}\lambda_2^4$	$\frac{315i\beta_4}{128}\lambda_2^4$

 Table 16: Coefficients for $n = 3/2$, $m \in \{0, 1/2, 1, 3/2\}$.

Coeff.	$(3/2, 0)$	$(3/2, 1/2)$	$(3/2, 1)$	$(3/2, 3/2)$
C_1	0	$-\frac{5i\beta_4}{128}\lambda_2^4$	0	$\frac{3i\beta_4}{128}\lambda_2^4$
C_2	0	$\frac{\beta_3}{16}\lambda_2^3$	0	$-\frac{\beta_3}{16}\lambda_2^3$
C_3	0	$-\frac{i}{8}\Xi\lambda_2^2 + \frac{5i\beta_4}{96}\lambda_2^4$	0	$\frac{3i}{8}\Xi\lambda_2^2 - \frac{5i\beta_4}{32}\lambda_2^4$
C_4	0	$\frac{1}{2}\lambda_1 - \frac{11\beta_3}{48}\lambda_2^3$	$\lambda_1 - \frac{7\beta_3}{12}\lambda_2^3$	$\frac{3}{2}\lambda_1 - \frac{17\beta_3}{16}\lambda_2^3$
C_5	$\frac{g_0-\alpha}{2} - \frac{3i}{4}\Xi\lambda_2^2 + \frac{13i\beta_4}{32}\lambda_2^4$	$\frac{g_0-\alpha}{2} - \frac{5i}{4}\Xi\lambda_2^2 + \frac{163i\beta_4}{192}\lambda_2^4$	$\frac{g_0-\alpha}{2} - \frac{7i}{4}\Xi\lambda_2^2 + \frac{139i\beta_4}{96}\lambda_2^4$	$\frac{g_0-\alpha}{2} - \frac{9i}{4}\Xi\lambda_2^2 + \frac{141i\beta_4}{64}\lambda_2^4$
C_6	$-\frac{3}{2}\lambda_1 + \frac{13\beta_3}{8}\lambda_2^3$	$-\frac{3}{2}\lambda_1 + \frac{35\beta_3}{16}\lambda_2^3$	$-\frac{3}{2}\lambda_1 + \frac{11\beta_3}{4}\lambda_2^3$	$-\frac{3}{2}\lambda_1 + \frac{53\beta_3}{16}\lambda_2^3$
C_7	$\frac{15i}{8}\Xi\lambda_2^2 - \frac{85i\beta_4}{32}\lambda_2^4$	$\frac{15i}{8}\Xi\lambda_2^2 - \frac{105i\beta_4}{32}\lambda_2^4$	$\frac{15i}{8}\Xi\lambda_2^2 - \frac{125i\beta_4}{32}\lambda_2^4$	$\frac{15i}{8}\Xi\lambda_2^2 - \frac{145i\beta_4}{32}\lambda_2^4$
C_8	$-\frac{35\beta_3}{16}\lambda_2^3$	$-\frac{35\beta_3}{16}\lambda_2^3$	$-\frac{35\beta_3}{16}\lambda_2^3$	$-\frac{35\beta_3}{16}\lambda_2^3$
C_9	$\frac{315i\beta_4}{128}\lambda_2^4$	$\frac{315i\beta_4}{128}\lambda_2^4$	$\frac{315i\beta_4}{128}\lambda_2^4$	$\frac{315i\beta_4}{128}\lambda_2^4$

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